

Problem set 1

2026-02-02

Instructions/Setup

[Download source](#)

- Make sure you have a GitHub account. In your GitHub account, create a GitHub repository named STAT3000.
- Create an RStudio Project, e.g., `stat3000`, connected to the GitHub repo (clone or connect existing).
- Answer each item **inside this Quarto file**. Show commands/code where requested.

Tasks

0. Project folders

In your project root, create the folders: `hw/`, `img/`, `data/`, `code/`, `docs/`.

Provide either:

- the Unix commands you used, or
- the R commands you used

`# Linux command`

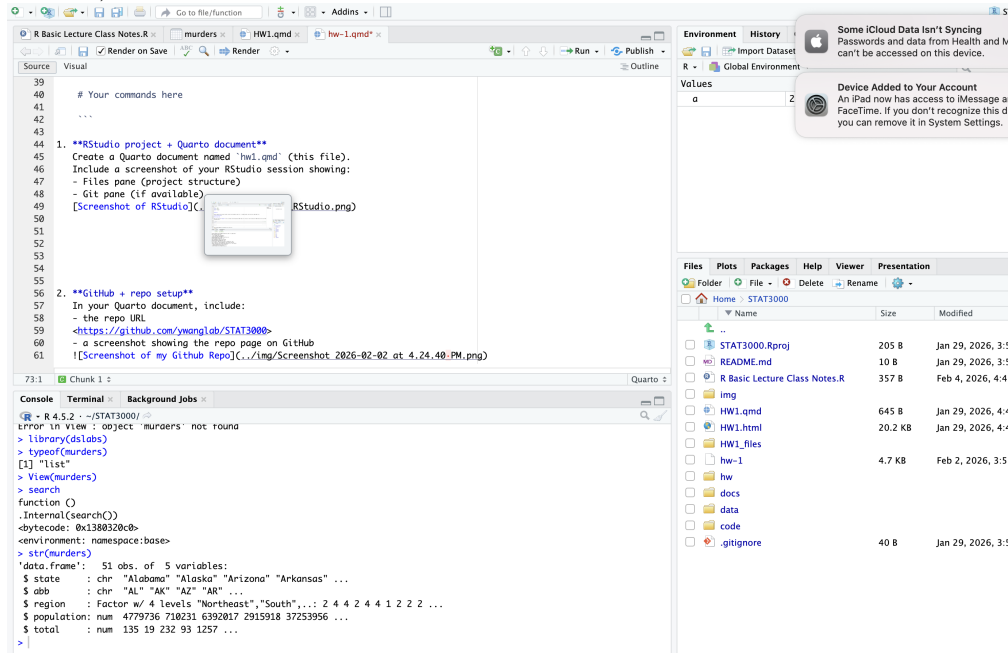
`# Your commands here`

```
mkdir hw
mkdir img
mkdir data
mkdir docs
```

1. RStudio project + Quarto document

Create a Quarto document named `hw1.qmd` (this file).
Include a screenshot of your RStudio session showing:

- Files pane (project structure)

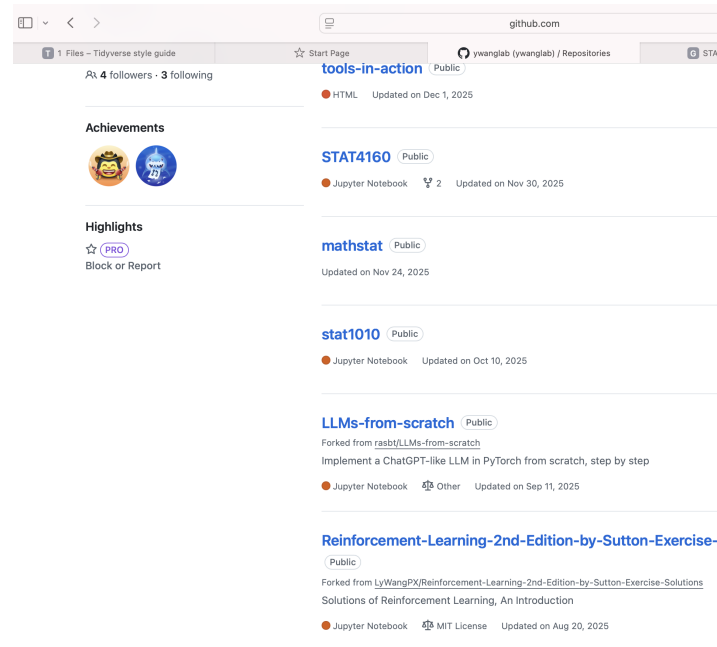


- Git pane (if available)

2. GitHub + repo setup

In your Quarto document, include:

- the repo URL <https://github.com/ywanglab/STAT3000>



- a screenshot showing the repo page on GitHub

3. Create and include a plot image

Create a simple plot in R (any plot you like), save it as `img/plot.png`, and include it below.

```
# Example idea (you can change it):
png("../img/cars.png", width = 800, height = 500) #open png device
plot(cars)
dev.off()
```

pdf
2

```
png("../img/plot.png", width = 800, height = 500)
plot(cars)
dev.off()
```

pdf
2

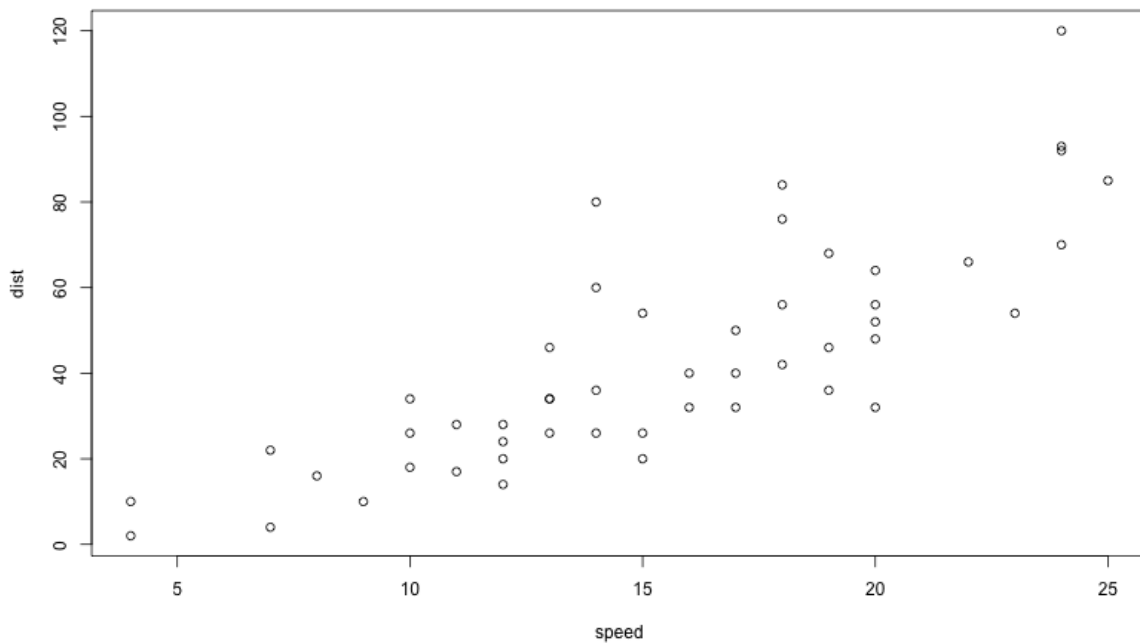


Figure 1: Simple cars plot

4. **Define a function and compute real roots** In this document, define coefficients $a = 2$, $b = -5$, $c = -3$. Consider $f(x) = ax^2 + bx + c$. Print the **real** solutions of $f(x) = 0$ (if any).

- If the discriminant is negative, print a message like "No real roots".

```
a <- 2
b <- -5
c <- -3
#
disc <- b^2 - 4*a*c
disc
```

[1] 49

5. **Graph the quadratic on a specified interval** Make a graph of $f(x)$ versus x for $x \in (-6, 6)$. Add:

```

# Create x values and compute f(x)
x <- seq(-6, 6, length.out = 400)
fx <- a * x^2 + b * x + c

```

```

# Coefficients

```

```

a <- 2
b <- -5
c <- -3

```

```

# Create x values and compute f(x)

```

```

x <- seq(-6, 6, length.out = 400)
fx <- a * x^2 + b * x + c

```

```

# Plot f(x)

```

```

plot(x, fx, type = "l",
      xlab = "x", ylab = "f(x)",
      main = "Graph of f(x)")

```

```

# Horizontal line at y = 0

```

```

abline(h = 0, col = "gray", lty = 2)

```

```

# Compute discriminant

```

```

disc <- b^2 - 4 * a * c

```

```

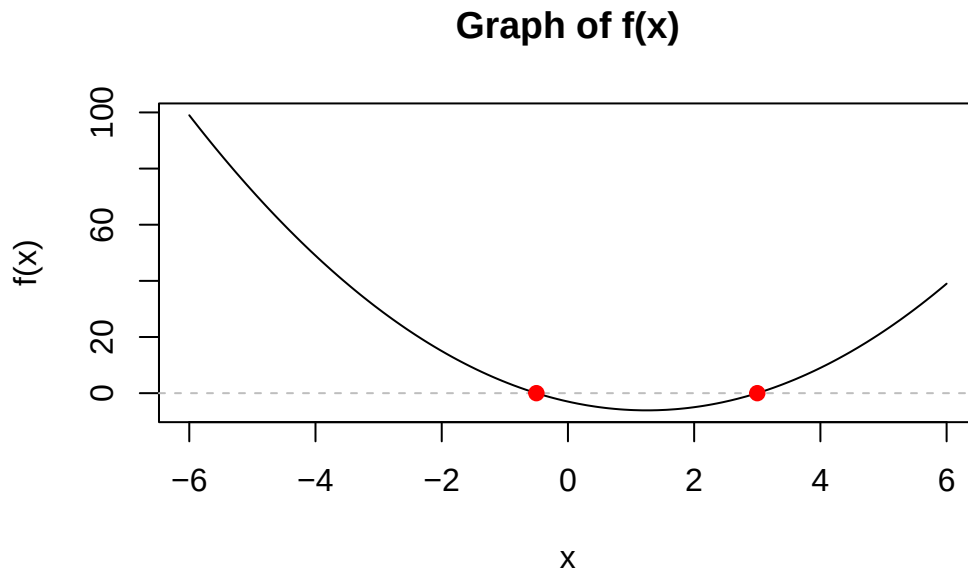
# Plot real roots if they exist

```

```

if (disc >= 0)
# roots <- # Plot real roots if they exist
if (disc >= 0) {
  roots <- c((-b + sqrt(disc)) / (2*a),
            (-b - sqrt(disc)) / (2*a))
  points(roots, rep(0, length(roots)), col = "red", pch = 19)
}

```



```
# x <- seq(-6, 6, length.out = 400)
# fx <- a*x^2 + b*x + c

# Hint: use plot(x, fx, type="l") and abline(h=0)
```

6. **Write coefficients to a text file using Unix** Use Unix to create a file `data/coeffs.txt` containing a single line with: `mkdir -p data echo "2 -5 -3" > data/coeffs.txt`

2 -5 -3

Show the Unix commands you used:

```
mkdir -p data
echo "2 -5 -3" > data/coeffs.txt
```

7. **Copy the Quarto file using Unix** Use Unix to copy `hw1.qmd` into `code/` and name it `quadratic.qmd`. Show the Unix command(s):

```
mkdir -p code
cp hw1.qmd code/quadratic.qmd
```

8. **Read coefficients from file using a relative path**

Edit `code/quadratic.qmd` so it reads `a`, `b`, `c` from `data/coeffs.txt` using a **relative path**. In *this* document, show the R code you used (it should also appear in `quadratic.qmd`).

```

coefs <- scan("../data/coefs.txt")

a <- coefs[1]
b <- coefs[2]
c <- coefs[3]

disc <- b^2 - 4*a*c
disc

```

```
[1] 49
```

9. Switch to an absolute path and test portability

In `code/quadratic.qmd`, replace the relative path with: `file.path(getwd(), "data/coefs.txt")` Render it once to confirm it works.

Then **move the entire project folder** to a new folder name (e.g., `RtmpXXXX`). Re-render `code/quadratic.qmd`. Does it still render? Briefly explain what happened, then switch back to a relative path and confirm it renders again.

10. Render to PDF and publish to GitHub Pages-style docs folder

Render a PDF version of `hw1.qmd` (or `code/quadratic.qmd`) and place the PDF into `docs/`. Show the terminal command(s) you typed:

```
quarto render hw/hw-1.qmd
```

11. Git workflow

Make at least **three commits** with meaningful messages, such as:

* Add initial project structure with folders `img`, `data`, `code`, `docs`

```

git add -A
git commit -m "Add initial project structure"
git push

```

* Implement quadratic discriminant and root calculations in `quadratic.qmd`

```

git add .
git commit -m "Implement quadratic discriminant and root calculations"
git push

```

* Render project output to `docs` folder and update repository

```

git add docs/
git commit -m "Render PDF output to docs folder"
git push

```

In this document, paste the output of:

```
2bbea6c (HEAD -> main, origin/main, origin/HEAD) Add initial project structure
cf6b925 Add initial project structure
7584aec my second commit
3d17757 update docs
6dc6c65 my first commit
```

```
git log --oneline --decorate -n 5
```

12. **Push to GitHub** Push your work to GitHub. In this document, include:

- a screenshot of your GitHub repo showing the folders `img/`, `data/`, `code/`, `docs/`

 **tfelton5** Add initial project structure ✓

2bbea6c · 22 minutes ago 6 Commits

HW1_files	Add initial projec...	22 minutes ago
code	Add initial projec...	22 minutes ago
data	Add initial projec...	22 minutes ago
docs	Add initial projec...	22 minutes ago
hw	Add initial projec...	22 minutes ago
img	Add initial projec...	22 minutes ago
.DS_Store	Add initial projec...	22 minutes ago
.gitignore	Add initial projec...	22 minutes ago
HW1.html	my second commit	2 weeks ago
HW1.qmd	my second commit	2 weeks ago
R Basic Le...	Add initial projec...	22 minutes ago
README....	Initial commit	2 weeks ago
STAT3000...	my first commit	2 weeks ago
hw-1	Add initial projec...	2 weeks ago
hw1.tex	Add initial projec...	22 minutes ago

 **README** 

STAT3000

- the link to the rendered PDF file in your repo (if applicable) <https://github.com/tfelton5/STAT3000/blob/main/docs/hw-1.pdf> —

Submission checklist

- ☐ Repo exists and is named HW1
- ☐ `hw1.qmd` renders without errors
- ☐ `img/plot.png` exists and is included in the document
- ☐ `data/coeffs.txt` exists and is read using a relative path
- ☐ `docs/` contains the rendered PDF
- ☐ At least three commits + pushed to GitHub